Pick a paper

Each paper below cites PaintOmics and used it on this data. We ran the same data on paintomics.org, and PaintOmics AI wrote what it found as numbered statements. Pick a paper, open its tab in the marking sheet and give every statement one of four marks: makes sense, does not make sense, especially good, can't judge. The job page shows the evidence behind each statement.

A Hernandez 2017 Mol Neurodegener

Does deleting the C5a receptor change microglia in an Alzheimer's model?

Hernandez MX, Jiang S, Cole TA, et al. Prevention of C5aR1 signaling delays microglial inflammatory polarization, favors clearance pathways and suppresses cognitive loss. Mol Neurodegener. 2017;12(1):66.

Mouse · microglia · Arctic model × C5ar1 knockout · 2, 5, 7, 10 months

Gene expression (RNA-seq time course)

In the paper: PaintOmics 3: KEGG enrichment of each of nine time-course gene clusters (Additional file 7), used to choose the genes of Figs. 5 and 6.

The paper's main conclusions

1. Inflammatory genes (NF-kappa B, cytokine-cytokine receptor, Toll-like receptor and chemokine signalling) rise in Arctic microglia from 5 months and stay lower when C5ar1 is deleted.
2. Phagosome, lysosome and antigen-presentation genes are increased in Arctic microglia and increased further in Arctic/C5aR1KO, from as early as 2 months.
3. Deleting C5ar1 prevents the spatial-memory deficit and the loss of neuronal complexity at 10 months without changing plaque load (behaviour and histology, not in the omics).

the job

the paper

`7NHQTK7rWz`

B Wang 2019 iScience

What does HNF4α need to make hepatocytes? Two edited mutants

Wang Y, Tatham MH, Schmidt-Heck W, et al. Multiomics Analyses of HNF4alpha Protein Domain Function during Human Pluripotent Stem Cell Differentiation. iScience. 2019;16:206-217.

Human · stem-cell-derived hepatic progenitors (day 9) · HNF4A DNA-binding-domain deletion and SUMO-site mutant vs wild type

Gene expression + proteomics

In the paper: PaintOmics 3: joint transcriptome + proteome KEGG enrichment of each mutant (section "Integrating Transcriptomics and Proteomics Datasets", Fig. 6A).

The paper's main conclusions

1. HNF4α without its DNA-binding domain cannot specify hepatic progenitors; the SUMO-site point mutant makes progenitors but fails to form functional hepatocytes (phenotype, not in the omics).
2. Both mutants share dysregulated genes and proteins in cholesterol/lipid metabolism and cell-matrix interaction: cholesterol metabolism is the top KEGG pathway in the DBD mutant, and five of its top-10 pathways (cholesterol metabolism, complement and coagulation, ECM-receptor interaction, focal adhesion, cell adhesion molecules) are shared with the SUMO mutant.
3. The DBD mutant also loses mitochondrial proteins: subunits of respiratory-chain complexes I to V are down (proteomics).

the job

the paper

`Y6n7275a0Z`

Marking sheet: https://docs.google.com/spreadsheets/d/1iW8rkhe8KT5AWozriMbvSKxYZA8Xa6lObqIG04MJPTw/edit?usp=sharing
Slides and materials: github.com/ConesaLab/SEBiBC2026

Pick a paper

C Jiménez-Munguía 2021 Sci Rep

How do brain endothelial cells respond to pneumococcus and to its adhesin alone?

Jiménez-Munguía I, Tomečková Z, Mochnáčová E, et al. Transcriptomic analysis of human brain microvascular endothelial cells exposed to laminin binding protein (adhesion lipoprotein) and Streptococcus pneumoniae. Sci Rep 11:7970 (2021).

Human · brain microvascular endothelial cells · live S. pneumoniae or its adhesion protein Lbp vs untreated cells

Gene expression (RNA-seq)

In the paper: PaintOmics: KEGG pathways and molecular functions of the genes responding to both treatments (Table 4, Fig. 2A, Suppl. dataset 1.8).

The paper's main conclusions

1. The adhesin alone reproduces much of the response to the live bacterium: 350 genes change after S. pneumoniae and 443 after Lbp, 231 of them in both.
2. The shared response is inflammatory signalling: the common genes take part in cytokine and chemokine signalling, TNF signalling and phagosome formation.
3. Lbp additionally induces cell senescence and breakdown, and remodelling of the extracellular matrix (Reactome analysis).

the job

the paper

`2Dh754YEIO`

D Balzano-Nogueira 2021 Genome Biol

What changes in blood in the year before islet autoimmunity?

Balzano-Nogueira L, Ramirez R, Zamkovaya T, et al. Integrative analyses of TEDDY Omics data reveal lipid metabolism abnormalities, increased intracellular ROS and heightened inflammation prior to autoimmunity for type 1 diabetes. Genome Biol 22:39 (2021).

Human · TEDDY birth cohort · children who later develop islet autoantibodies vs matched controls · 12, 9, 6, 3 and 0 months before seroconversion

Gene expression + metabolomics (with lipid classes)

In the paper: PaintOmics 3: the predictive signature projected onto KEGG pathway maps, from which the metabolic and immune models of Figs. 5 and 6 were drawn.

The paper's main conclusions

1. A multi-omics signature (862 genes, 245 metabolites, 3 dietary biomarkers) separates children who will develop islet autoimmunity from matched controls up to 12 months before seroconversion; the 12- and 9-month windows carry most of the information.
2. Cases show a sustained metabolic phenotype: impaired lipid metabolism (structural lipids and the APOA1/PPARα axis down) with glycolysis, TCA cycle and oxidative phosphorylation up, leading to accumulation of reactive oxygen species.
3. That state is linked to inflammation and immune activation before any autoantibody appears: arachidonate metabolism, early TNFα, antigen presentation, low IL10/TGFβ, matrix remodelling and cytotoxicity genes.

Held back on the job page by the "not a graph artifact" check: click "Show the interpretation anyway" there. The marking sheet lists its statements in full.

Ana Conesa is the senior author.

the job

the paper

`560S30473C`

Marking sheet: **https://docs.google.com/spreadsheets/d/1iW8rkhe8KT5AWozriMbvSKxYZA8Xa6lObqIG04MJPTw/edit?usp=sharing**
Slides and materials: **github.com/ConesaLab/SEBiBC2026**

Pick a paper

E Rubio 2021 Sci Rep

Minimal hepatic encephalopathy in cirrhosis: what changes in blood?

Rubio T, Felipo V, Tarazona S, et al. Multi-omic analysis unveils biological pathways in peripheral immune system associated to minimal hepatic encephalopathy appearance in cirrhotic patients. Sci Rep 11:1907 (2021).

Human · patients with vs without MHE · blood cells and serum

Gene expression + metabolomics + serum cytokines

In the paper: PaintOmics 3: KEGG enrichment of the differential genes and the network of significant pathways (Fig. 2, Table S2).

The paper's main conclusions

1. Blood-cell gene expression points to a shift in T-helper populations: Th1/Th2 and Th17 differentiation, Jak-STAT, cytokine-cytokine receptor interaction and cell adhesion molecules form one network of altered immune pathways.
2. A second group of altered pathways is lipid metabolism: fat digestion and absorption, cholesterol metabolism, steroid and primary bile acid biosynthesis, PPAR signalling.
3. Serum molecules link to the transcriptome in modules: CCL20, CX3CL1, CXCL13, IL-15, IL-22 and IL-6 with altered chemotaxis; long-chain unsaturated phospholipids with fatty-acid transport and prostaglandin production.

Held back on the job page by the "not a graph artifact" check: click "Show the interpretation anyway" there. The marking sheet lists its statements in full.

Ana Conesa and Sonia Tarazona are co-authors.

the job

the paper

`3oP1p37B25`

F Fàbregas 2018 Nat Commun

Why do plants that overexpress the BRL3 receptor survive drought?

Fabregas N, Lozano-Elena F, Blasco-Escamez D, et al. Overexpression of the vascular brassinosteroid receptor BRL3 confers drought resistance without penalizing plant growth. Nat Commun. 2018;9:4680.

Arabidopsis · roots · BRL3-overexpressing vs wild type, well watered and after 5 days without water

Gene expression + metabolomics

In the paper: PaintOmics 3: joint KEGG enrichment of root transcripts and metabolites, run for watered and for drought (Suppl. Fig. 11, Suppl. Data 16 and 17).

The paper's main conclusions

1. BRL3-overexpressing roots carry a primed osmoprotectant signature: proline and sugars (sucrose, trehalose, myo-inositol) are already higher when well watered, and trehalose, sucrose, proline and raffinose rise further under drought.
2. Stress-response genes (water deprivation, abscisic acid) are constitutively active in BRL3-overexpressing roots, and the deregulated genes are enriched in the root vasculature.
3. Joint transcript and metabolite pathway analysis points to sugars: sucrose metabolism is deregulated constitutively and more so under drought, and galactose metabolism (raffinose-family oligosaccharides) under drought; BRL3 promotes drought tolerance mainly by controlling sugar metabolism.

Held back on the job page by the "not a graph artifact" check: click "Show the interpretation anyway" there. The marking sheet lists its statements in full.

From CRAG, Barcelona. Ana Conesa is a co-author.

the job

the paper

`044DBhQFC7`

Marking sheet: **https://docs.google.com/spreadsheets/d/1iW8rkhe8KT5AWozriMbvSKxYZA8Xa6lObqIG04MJPTw/edit?usp=sharing**
Slides and materials: **github.com/ConesaLab/SEBiBC2026**